

$^{33}\text{S}(^3\text{He},n\gamma)$ 1975Da14

$^{33}\text{S}+2p \rightarrow ^{35}\text{Ar}$ from $J^\pi=3/2^+$ ^{33}S ground state.

1975Da14: a $^3\text{He}^+$ beam was produced from the University of Alberta Van de Graaff accelerator. Targets were $150\text{ }\mu\text{g}/\text{cm}^2$ layers of Ag_2S (59% ^{33}S) on silver backings. At $E(^3\text{He})=6.375\text{ MeV}$, neutrons were detected using an NE213 liquid scintillator placed at $\theta_{\text{lab}}=0^\circ$, 10° , and 20° . At $E(^3\text{He})=6.660\text{ MeV}$, neutrons were detected using the NE213 scintillator placed at $\theta_{\text{lab}}=0^\circ$ and 20° . At $E(^3\text{He})=6.390\text{ MeV}$, neutrons were detected using the NE213 scintillator placed at $\theta_{\text{lab}}=0^\circ$ and neutron-coincident γ rays were detected using a 10% efficient Ge(Li) detector placed at 90° . Measured time-of-flight (TOF) spectra of neutrons, $\sigma(E_n, \theta)$, E_γ , $n\gamma$ -coin. Deduced Q values, levels, L_{2p} , J, π , and isospin.

 ^{35}Ar Levels

$E(\text{level})^\dagger$	J^π	$L^\ddagger$	Comments
0	$3/2^+$	(0)	J^π : mirror $3/2^+$ ^{35}Cl g.s.
1184.2 6			
1749.8 9			
2600.8 15			J^π : 1975Da14 observed $L(^3\text{He},n)=(0)$ for a 2.60-MeV level in ^{35}Ar , implying the existence of a $3/2^+$ level, which possibly corresponds to the $3/2^+$ 2638-keV level in ^{35}Ar .
3195.8 11			E(level): deduced by evaluators from the unplaced 1446.0 γ .
5537 25	$(3/2^+)$	(0)	T=3/2 E(level): from measured neutron energy corresponding to this level. J^π : $(\pi 1d_{3/2})^2(\nu 1d_{3/2})^1$ configuration formed by the $J^\pi=0^+$, T=1 $(\pi 1d_{3/2})^2$ di-proton transfer from ^3He to ^{33}S of $^{32}\text{S}(\nu 1d_{3/2})^1$ configuration. Tentative first T=3/2 state in ^{35}Ar .

† From **1975Da14** based on measured E_γ which however are not listed by the authors, unless otherwise noted.

‡ the observed maximum at $\theta=0^\circ$ in $\sigma(E_n, \theta)$ implies $L=0$.

 $\gamma(^{35}\text{Ar})$

$E_\gamma^\dagger$	$E_i(\text{level})$	E_f	J_f^π	Comments
1184.2	1184.2	0	$3/2^+$	
1446.0 6	3195.8	1749.8		Unplaced γ in 1975Da14 . Evaluators placed it based on the Adopted Levels from $^{16}\text{O}(^{24}\text{Mg}, \alpha n\gamma)^{35}\text{Ar}$ (2004Ek01) and $^{24}\text{Mg}(^{16}\text{O}, \alpha n\gamma)^{35}\text{Ar}$ (2007De14).
1749.8	1749.8	0	$3/2^+$	
2600.7	2600.8	0	$3/2^+$	

† E_γ values without uncertainties are deduced by evaluators from the level-energy differences from **1975Da14**.

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Level Scheme

